MILITARY SPECIFICATION

FEED MECHANISMS, 20MM, MARK 7 MODS 0 AND 2

1. SCOPE

1.1 Scope. - This specification covers the procurement requirements for 20MM Feed Mechanisms Mark 7 Mods 0 and 2.

1.2 Classification. - Feed mechanisms furnished under this specification shall be of the following Mark and Mods, as specified (see 6.2):

Mark 7 Mod 0 (LD 276462)
Mark 7 Mod 2 (LD 266061)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, together with those listed on applicable drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

P-S-566 - Soap, Chip.
P-S-661 - Solvent, Dry Cleaning.

MILITARY

JAN-C-372 - Cleaner, Rifle-Bore.
MIL-I-6870 - Inspection Requirements, Non-destructive; for Aircraft Materials and Parts.
MIL-R-15074 - Compound, Corrosion Preventive, Fingerprint Remover.
MIL-L-19701(NOrd) - Lubricant, All Weather, Semi-Fluid, for Aircraft Ordnance.

FED. SPEC. CLASS. 1005

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STANDARDS

MILITARY

MIL-STD-10 - Surface Roughness, Waviness, and Lay.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

BUREAU OF ORDNANCE

NAVORD OSTD 2 - Method of Marking Ordnance Material.

DRAWINGS

BUREAU OF ORDNANCE

982519 - 20MM Ammunition for Aircraft Gun Mk 12, Target Practice Round.
982520 - 20MM Ammunition for Aircraft Gun Mk 12, Dummy Round.
1445196 - 20MM Dummy Round, for Aircraft Gun Mk 12.
1500629 - Power Supply Console (for testing 20MM Machine Gun Mk 12 Mods).
1515855 - General Arrangement Wiring Diagram, Power Supply Console (for testing 20MM Machine Gun Mk 12 and Mods).
1591132 - Wiring Diagram, Standard Proofing Equip.
LD 172560 - Synchronizing Switch Mk 1 Mod 1.
LD 255941 - 20MM Ammunition Link Mk 2 Mod 0.
LD 256043 - 20MM Ammunition Link Mk 2 Mod 1.
LD 256076 - Shipping Container for 20MM Feed Mechanism Mark 7 Mods 0 and 2.
LD 266032 - 20MM Aircraft Gun Mk 12 Mod 0.
LD 266061 - 20MM Feed Mech. Mk 7 Mod 2.
LD 276462 - 20MM Feed Mech. Mk 7 Mod 0.
LD 288830 - 20MM Ammunition Link Mk 2 Mod 2.
LD 413813 - Power Supply Console (For Testing 20MM Machine Gun Mk 12 and Mods).
LD 413815 - Standard Proofing Equip. (for 20MM Gun Mk 12 and 20MM Feed Mech. Mk 7 & Mk 8).
LD 413816 - Fixed Rigidity Adapter Assembly.
LD 414203 - Electrical Equipment (for Power Supply Console for Testing 20MM Aircraft Gun Mk 12 Mods).

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SK. 234828 - Gun Mount Assembly (20° Depression).
SK. 234831 - Gun Mount Assembly (Horizontal).
SK. 401113 - Link Chute (L.H.) Fabrication.
SK. 401119 - Link Chute (R.H.) Fabrication.
SK. 407729 - 20MM Feed Mechanism Mark 7 Mods, Pneumatic Leakage Test Set Schematic.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Material.
OP 1912 - 20MM Feed Mechanism Mark 7 Mods 0 and 2 Instruction Manual.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services (with Supplements).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction sample. - Unless otherwise specified in the contract or order (see 6.2), preproduction sample (see 4.1.1) of feed mechanisms are required. The samples shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in 4.3.1 and are for the purpose of determining that the manufacturers production process will produce feed mechanisms that will meet the requirements of this specification.

3.2 General. - General requirements shall be in accordance with Publication OP 400. The design characteristics, mechanical requirements, and the performance of the feed mechanisms covered by this specification shall conform to the applicable drawings and this specification.

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3.2.1 Contractor obligation. - Means and methods of manufacture will be determined by the contractor; however, the obligation is on the contractor to satisfy the procuring activity as to full compliance with the contract, the applicable drawings as listed on LD 266061 or LD 276462, and this specification. The contractor shall provide adequate safety devices to prevent injury to personnel during the tests of 4.3.

3.3 Material. - Material shall conform to the requirements and specifications specified on the applicable drawings. In procuring materials the contractor shall provide for material lot formation as specified in 4.1.2.1 and 4.1.2.2.

3.4 Physical requirements. - The physical requirements of the component parts of the feed mechanisms covered by this specification shall be in strict accordance with the applicable drawings and this specification. In addition, the component parts shall be free from cracks, porosity, voids, scale, or other defects which may make the part unsuitable for the intended purpose.

3.5 Finishes, coatings and tolerances. -

3.5.1 Surface roughness, waviness, and lay. - Corners, edges, and surfaces shall be finished in strict accordance with drawing requirements and shall be free from any roughness, tool marks, machining mismatches, or any other discrepancies which would tend to initiate cracks or cause galling or other malfunction. Surface roughness symbols shown on the drawings shall be interpreted in accordance with Standard MIL-STD-10.

3.5.2 Protective coating. - Unless otherwise specified in the contract or order, all parts of the feed mechanisms covered by this specification shall be plated, coated, or treated as specified on the applicable drawings.

3.5.3 Dimensions and tolerances. - Dimensions and tolerances shall be in strict accordance with the applicable drawings and this specification.

3.6 Interchangeability. - All parts having the same part number shall be completely interchangeable with each other and with like parts of other manufacture with respect to installation and performance. Parts joined together by welding, brazing, or riveting shall be interchangeable as assemblies.

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3.7 Spare parts. - When procurement documents (see 6.2) stipulate that spare parts are to be furnished with assembled feed mechanism, the spare parts shall be produced concurrently with and of the same material lots as like parts used in the assembled feed mechanisms.

3.8 Threads. - Unless otherwise specified on the applicable drawings, all threads shall conform to the National Bureau of Standards Handbook H28, and the not-go practice shall be as set forth in the supplement to the aforementioned handbook. The class of fit shall be as specified on the applicable drawings.

3.9 Gages. - The contractor shall provide himself with whatever gages are necessary to insure that the parts and assemblies covered by this specification will meet the dimensions and tolerances specified by the applicable drawings. If so desired, and upon request of the contractor, the Government will furnish available drawings of the applicable Government final inspection gages required by the contractor. This procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such other gages as may be required to satisfactorily fulfill the contract requirements.

3.10 Identification marking. - Each component of the feed mechanism shall be marked in accordance with the drawings and Standards NAVORD OSTD 2 and MIL-STD-130 except that marking with ink on metal parts will not be permitted.

3.11 Lubrication. - Feed mechanisms shall be lubricated in accordance with Publication OP 1912 with a light coating of MIL-L-19701 (NOrd).

3.12 Assembly. - Prior to bench tests, cycling tests and firing tests, the feed mechanisms shall be assembled to the hand of feed specified (see 6.2) and serviced in accordance with applicable drawings and Publication OP 1912. There shall be no fitting or alteration of any kind during assembly or operation.

3.13 Government loaned property. - The contractor shall be responsible for any damage to Government loaned property. The contractor shall not machine, file, grind, scrape or otherwise modify such property except with the prior approval of the procuring activity.

3.14 Government furnished property. - The Government shall furnish any components when specified in the contract or order (see 6.2). Immediately upon delivery of Government furnished property, the contractor shall determine whether or not such property is defective or damaged. If such property is found to

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be defective or damaged, the contractor shall notify the Government and the Government shall repair or replace defective or damaged property at no expense to the contractor. However, if the Government furnished property is found to be in good condition and is accepted by the contractor and is subsequently damaged or after acceptance by the contractor is found to be defective, the responsibility for repairs or replacement shall rest with the contractor.

3.15 Performance. - Feed mechanisms shall perform satisfactorily when submitted to the tests described in 4.3.2, 4.3.3, 4.3.4, and 4.3.5. Evidence of any condition which indicates a tendency to gall or bind, or adversely affects rapidity and smoothness of function, or causes undue wear, or indicates the presence of burrs, leakage of air, or other defects, shall be considered as reason for rejection. The contractor shall correct, without additional expense to the Government, all defects due to improper adjustment, faulty material, or inferior workmanship.

3.15.1 Rate of fire. - The rate of fire (cycle) of the feed mechanism shall be not less than 925 nor more than 1125 rounds per minute.

3.16 Supplementary requirements. -

3.16.1 Reports of tests. - Accurate reports of parts failure, functional failure, and parts replacement of each feed mechanism, together with the applicable number of rounds fired shall be recorded. This information, together with an analysis of any malfunction shall be forwarded to the procuring activity.

3.17 Workmanship. - The quality of workmanship shall be such as to assure compliance with all the requirements of the applicable drawings, standards, and specifications.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS.

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of Standard MIL-STD-105.

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4.1.1 Preproduction sample. - Unless otherwise specified (see 4.1.1.1), preproduction samples manufactured in accordance with 3.1 shall consist of 3 left hand and 3 right hand assembled feed mechanisms Mark 7 Mod 0 or Mod 2 as specified (see 6.2). The sample shall be submitted to the Government inspector for preproduction testing at an activity designated by the procuring activity, for purposes of determining compliance with the requirements of the contract or order, and the applicable specifications, and drawings. Further production of the feed mechanisms or parts by the contractor prior to the approval of the testing or procuring activity (see 4.3.5.4.1) or completion of inspection and tests on the preproduction sample shall be at the contractors risk. Accepted preproduction samples will become the property of the procuring activity and will be included in the quantity of feed mechanisms called for in the contract or order.

4.1.1.1 Preproduction sample for a subsequent contract. - If a contractor has previously delivered feed mechanisms in accordance with the requirements of this specification and this product has been found to be satisfactory, the preproduction sample for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.2).

4.1.2 Lots. - Materials, assembled feed mechanisms, and spare parts shall be submitted for inspection in lots as follows:

4.1.2.1 Material lot. - A material lot shall consist of all raw material of the same kind, or origin, melt or heat, procured and processed into production at the same time, for manufacture of feed mechanisms and spare parts. Each such lot shall be submitted for inspection as prescribed in the applicable material specification. It shall be arranged in such a manner as to satisfy the inspector that all material in the lot is reasonably uniform in quality and characteristics.

4.1.2.2 Feed mechanisms and spare part lot. - A feed mechanism and spare part lot shall consist of assembled feed mechanisms and their spare parts as produced concurrently by a manufacturing process common to the lot. Each lot of assembled feed mechanisms and spare parts shall not include more than one lot of raw material where parts are subjected to heat treatment. Unless otherwise specified, the number of feed mechanisms and their spare parts in an "inspection lot" shall not exceed one (1) month's production.

4.1.2.3 Replacement parts lot. - A replacement parts lot shall consist of all spare parts of the same design presented for acceptance at one time, in which the unit of product is a part manufactured separate from the process of assembled feed mechanism lots (see 4.1.2.2).

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4.1.2.4 Lot identification. - The contractor shall mark the lot number and when applicable the heat, melt, or other manufacturing process numbers, directly on the material or parts or containers, in a manner approved by the inspector. Such identifications shall be maintained until the material or parts have been accepted. Where identification of raw material by lot number is necessary in the manufacture of parts, such identification shall be maintained during fabrication of parts.

4.2 Inspection. -

4.2.1 General. - The general inspection and test procedures for contractor inspection and Government inspection shall be in accordance with the provisions of Publication OP 400 and this specification. The inspection level, Acceptable Quality Level (AQL), and Ordnance Classification of Defects shall be provided by the procuring activity.

4.2.2 Inspection and test procedures. - Raw materials shall be tested for compliance with the applicable material specification. Parts and assembled feed mechanisms, spare parts, and replacement parts selected in accordance with 4.1 shall be visually and dimensionally inspected and tested for mechanical properties and material soundness in accordance with the Ordnance Classification of Defects, the applicable drawings, and this specification. The tests of 4.3.2, 4.3.3, and 4.3.4 shall be conducted by the contractor at the place of manufacture. The tests of 4.3.5.5 and 4.3.5.6 shall be conducted by the Government inspector at the testing activity designated by the procuring activity (see 6.2). Each preproduction sample (see 4.1.1) shall be subjected to all the inspections and tests of this specification.

4.3 Tests. -

4.3.1 General. - The assembled feed mechanisms comprising each inspection lot (see 4.1.2.2) and each preproduction sample (see 4.1.1) shall be subjected to bench tests of 4.3.2, functional cycling tests of 4.3.3, interchangeability tests of 4.3.4, firing tests of 4.3.5, and post proof visual inspection of 4.4. Spare parts and replacement parts shall be tested in accordance with 4.2.2 and shall meet the acceptance criteria of 4.5.6 or 4.5.6.1 whichever is applicable.

4.3.2 Bench tests. - Each feed mechanism of the inspection lot shall be subjected to the tests of 4.3.2.1 thru 4.3.2.12 and to 4.3.2.13 as applicable.

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4.3.2.1 The feed throat assembly shall be gaged to insure interchangeability of mounting on any receiver for the 20MM Aircraft Gun Mark 12.

4.3.2.2 The cover assembly shall be tested to assure that it latches properly to the feed throat assembly.

4.3.2.3 The cam-shaft-arm-pawl assembly group in the feed throat unit shall be tested for smooth operation without binding.

4.3.2.4 Ejectors shall be tested for clearance with the bolt body on the gun and smooth operation without binding.

4.3.2.5 The feed mechanism shall be tested to assure the proper attachment of adapters for both the feed chute and link chute.

4.3.2.6 Belt holding pawl assembly groups in the feed throat shall be tested for smooth operation without binding.

4.3.2.7 Drive pawls and slide groups and the last round lock shall be tested for smooth operation without binding in the cover drive assembly. The sear mechanism shall properly operate the valve as the drive pawl slide is moved to each end of its stroke.

4.3.2.8 The feed mechanism shall be tested to insure that standard attachments such as air line connector may be accommodated. The tie bar assembly shall be inspected and checked to assure proper engagement of the locking lugs with the end plate.

4.3.2.9 All screws, rivets, nuts, dowel pins, roll pins, and lock wire shall be inspected for proper assembly.

4.3.2.10 Feed mechanisms shall be inspected for proper lubrication.

4.3.2.11 Support pins in end plates shall be inspected for tightness.

4.3.2.12 Feed mechanisms shall be tested to determine the leakage of air from the feed mechanism. Air pressure shall be 800 pounds per square inch (psi) and the volume of entrapped air (external to the motor of the feed mechanism) shall not

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exceed 2 cubic inches (see SK 407729). Tests shall be conducted at the following stroke positions:

(a) When the last round lock is preventing motion of the drive slide on the return stroke.
(b) At the final stall position on the return stroke of the motor.

The loss of air shall not exceed 50 psi of entrapped air (including motor) in 5 minutes.

4.3.2.13 Feed mechanisms Mark 7 Mod 2. - Feed mechanisms Mark 7 Mod 2 shall be tested for continuity of circuit and resistance of circuit as follows:

(a) The circuit shall have a resistance of not more than 3 ohms with valve stem in, and a resistance of not less than 500 megohms with valve stem out.
(b) Resistance to ground shall be not less than 500 megohms.

4.3.3 Functional cycling tests. - Each feed mechanism of the inspection lot shall be tested by cycling at least fifteen rounds of dummy ammunition through the feed mechanism when mounted on a Government specified test set which simulates the action of the gun. Air pressure applied to the feed mechanism shall be between 550 and 570 psi at the intake to the motor under flow conditions using the equivalent of the piping arrangement shown on Drawing 1500629.

4.3.4 Interchangeability tests. - After the bench tests of 4.3.2 and the functional cycling tests of 4.3.3 have been completed on each lot, three right-hand and three left-hand feed mechanisms shall be disassembled and their parts intermingled (scrambled) with four sets of spare parts from each lot. This disassembly and scrambling of parts shall include all parts not permanently secured by brazing or riveting. After disassembly and scrambling, three right-hand and three left-hand feed mechanisms shall be assembled by selecting parts at random from the scrambled parts. There shall be no fitting or altering of any kind during the interchangeability tests. Preproduction samples may be tested without the use of spare parts.

4.3.4.1 All reassembled feed mechanisms of 4.3.4 shall receive the bench tests of 4.3.2 and the functional cycling tests of 4.3.3.

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4.3.5 Firing tests. -

4.3.5.1 Associated equipment. - The following associated equipment shall be used in the firing tests described in the following paragraphs:

(a) Gun Mount Assembly (Horizontal) Drawing SK 234831 or Gun Mount Assembly (20° depression) Drawing SK 234828 together with Fixed Rigidity Adapter Assembly LD 413816 or other mount approved by the procuring activity.
(b) 20MM Aircraft Gun Mark 12 Mod 0 (LD 266032) phased to fire 925 to 1125 rounds per minute - as determined by 4.3.5.2.
(c) 20MM Ammunition (Target Practice Round - Drawing 982519) for Mark 12 Mod 0 gun (steel cased).
(d) 20MM Ammunition (Inert) (Dummy Round Drawing 982520 or 1445196) for Mark 12 Mod 0 gun.
(e) 20MM Ammunition Link Mark 2 Mod 0 (LD 255941) or 20MM Ammunition Link Mark 2 Mod 2 (LD 288830) and 20MM Ammunition Link Mark 2 Mod 1 (End Link) (LD 256043).
(f) Flexible chute M12 - 110 (L.H.) and M12 - 111 (R.H.) with gun end for Mark 7 Feed Mechanisms M12 - 450 and Box end M12 - 320, Standard Armament Inc., or equivalent as approved by the procuring activity.
(g) Link chute SK 401119 R.H. and 401113 L.H. or equivalent as approved by the procuring activity.
(h) Synchronizing Switch Mark 1 Mod 1 (LD 172560) wired to fire through the slide contact of the gun.
(i) Pneumatic requirements as shown on Drawing 1500629 and LD 413813.
(j) Electrical firing circuit requirements as listed on Drawing 1515855 and LD 414203.
(k) Standard Proofing Equipment as shown on Drawing 1591132 and LD 413815.
(l) Other equipment as required to determine rate of fire.

4.3.5.2 Rate of fire. - The rate of fire shall be determined during the functional firing test for each feed mechanism fired, and shall be taken as the average rate determined for not less than 15 successive rounds within the last 20 rounds of a burst. The rate shall be determined using a 20MM Feed Mechanism Mark 7 Mod 0 or a 20MM Feed Mechanism Mark 7 Mod 2 with the phasing switch disconnected from the firing circuit. The rate of fire will decrease when the phasing switch is operating, but shall not drop below 925 rounds per minute in any test.

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4.3.5.3 Suspension of firing tests. - When determination for rejection (see 4.5) is reached during any test, all further tests of that lot will be suspended and the lot rejected.

4.3.5.4 Reports of tests. - Accurate reports of test results shall be recorded and forwarded as specified in 3.16.1.

4.3.5.4.1 Preproduction test report. - When preproduction inspection and tests are completed, the testing activity shall notify the Government inspector immediately, who in turn shall notify the contractor of the test results and of any specific deficiencies in his preproduction samples. The contractor shall take the necessary action to correct any deficiency, and may be required, at the discretion of the testing activity, to submit additional samples until the specified number of satisfactory samples have been submitted. Such samples shall be accompanied by a description of the changes made to correct the faults of the previous submissions.

4.3.5.5 Functional firing tests. - Each feed mechanism of the sample selected in accordance with 4.1 shall be installed on a gun with associated equipment as described in 4.3.5.1 and tested as follows:

4.3.5.5.1 Cycle the feed mechanism five times by actuating the charger. Use inert ammunition (dummy round).

4.3.5.5.2 Fire two twenty-five round continuous bursts. Follow with two twenty-five round intermittent bursts (two interruptions each burst). Ammunition belts shall consist of 25 live rounds followed by 15 dummy rounds.

4.3.5.5.3 After testing each sample mechanism as specified in 4.3.5.5.1 and 4.3.5.5.2, select five feed mechanisms from the sample tested and reassemble for opposite hand (left hand or right hand) and repeat tests of 4.3.5.5.1 and 4.3.5.5.2.

4.3.5.6 Endurance firing test. -

4.3.5.6.1 General. - Endurance firing tests are defined as those tests designed to test the durability of materials and quality of workmanship used in the manufacture of the lot. The testing activity shall be furnished with an applicable parts replacement schedule (see 6.2).

4.3.5.6.2 Endurance firing. - One feed mechanism from each inspection lot shall be installed on a gun with associated equipment as described in 4.3.5.1 and tested by firing 1000 rounds of ammunition in bursts of not more than 50 rounds. Each ammunition belt shall be terminated with 15 dummy rounds.

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Reassemble the feed mechanism for opposite hand feed and fire 1000 additional rounds. Rates of fire shall be taken at the beginning of the test, at 1000 rounds, and at the end of the test.

4.4 Post proof inspection and reconditioning procedures. -

4.4.1 Inspection. - The activity conducting the firing tests will disassemble each feed mechanism after firing and conduct the following post proof inspection on each feed mechanism assembly tested:

(a) Visually inspect all parts for the following:

1. Broken part
2. Cracked part
3. Galling of parts
4. Binding of parts
5. Loose or missing parts
6. Abnormal wear

(b) Magnetic particle inspect the following parts in accordance with Specification MIL-I-6870:

1. Stripper subassembly
2. Drive slide
3. Drive pawls
4. Sear slide
5. Spring housing
6. Ejectors

(c) Prepare report of tests in accordance with 3.16.1.

4.4.2 Reassembly. - The test activity shall clean (see 5.2) and reassemble each feed mechanism, replacing all broken or damaged parts. In the case of mechanisms that have been subjected to the endurance firing tests of 4.3.5.6, replace parts in accordance with the applicable parts replacement schedule if the logbook for any mechanism indicates that the schedule applies. The feed mechanism shall be cycled in accordance with 4.3.3 to assure proper reassembly and made ready for shipment in the manner outlined in Section 5.

4.5 Test acceptance criteria. -

4.5.1 Bench and functional cycling tests. - During the bench tests of 4.3.2 and functional cycling tests of 4.3.3, any failure shall be cause for rejection of the feed mechanism and exclusion from those feed mechanisms to be submitted to the Government inspector for acceptance inspection and test.

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4.5.2 Interchangeability test. - Failure of a feed mechanism to comply with any of the interchangeability tests of 4.3.4 shall be cause for rejection of the lot represented.

4.5.3 Functional and endurance firing tests. - During the functional firing tests of 4.3.5.5 and the endurance firing tests of 4.3.5.6, parts breakage or malfunctions caused by manufacturing deficiencies or use of improper materials in the feed mechanism shall be considered as cause for rejection of the entire lot represented. Repair or replacement of defective parts or feed mechanisms by the contractor at no cost to the Government will be required. The naval test activity or cognizant Government inspector will judge performance of each sample mechanism as satisfactory or unsatisfactory in accordance with 4.5.3.1.

4.5.3.1 Defects. - Assembled feed mechanisms or spare parts containing defects in excess of the following criteria shall be judged unsatisfactory. Defects shall be classified in accordance with 4.5.3.1.1 and 4.5.3.1.2 and acceptance or rejection of the inspection lot shall be determined in accordance with 4.5.3.2.

4.5.3.1.1 Critical defects. - Any failure of the feed mechanism assembly or parts to function properly, or any material failure, either of which may cause serious damage to the feed mechanism, or to the gun, or injury to personnel shall be classified as a critical defect. The following are examples of critical defects:

(a) Explosion of a cartridge during feeding.
(b) Debulleting during feeding.
(c) Tearing cartridge case to the extent powder spills.

4.5.3.1.2 Major defects. - Any failure of the feed mechanism assembly or parts to function properly, or any material failure, either of which indicates that the feed mechanism is unsuitable for its intended purpose, shall be classified as a major defect. The following are examples of major defects:

(a) Part breakage.
(b) Cracked part.
(c) Galling of parts.
(d) Binding of parts.
(e) Loose or missing parts.
(f) Air leaks.
(g) Electrical failures.
(h) Abnormal wear.

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4.5.3.2 Lot acceptance for bench, cycling, and functional firing tests. - The lot will be accepted on the following basis:

(a) Functional firing tests - No critical failures in firing tests of the sample.
(b) Functional firing tests - Lot acceptance for major defects shall be judged in accordance with Standard MIL-STD-105, AQL 1.0 percent.
(c) Failures are not of a recurring type. If the same failure occurs more than once, the lot shall be rejected.

Failure of the test samples to meet the acceptance criteria of (a), (b), and (c), shall be cause for rejection of the entire lot.

4.5.4 Lot acceptance for endurance firing test. - During the endurance firing tests, one major failure but no critical failure attributable to the feed mechanism will be allowed for each 1000 rounds fired. Failures occurring in excess of the above allowance shall be cause for rejection of the entire lot represented. Feed mechanisms satisfactorily completing the endurance test will be issued to service after reconditioning.

4.5.5 Undetermined failures. - Any feed mechanism which fails the firing tests will be disassembled and reinspected for all characteristics which could contribute to the malfunction. If the cause of the malfunction cannot be determined and the associated equipment of 4.3.5.1 is determined to be satisfactory the feed mechanism will be retested and the first malfunction will not be counted against the feed mechanism or the lot.

4.5.5.1 Failures not attributable to feed mechanisms. - Failures occurring from causes not attributable to the feed mechanism or to manufacturing deficiencies will not be considered as a defect or cause for rejection, but shall be recorded and reported to the procuring activity. The tests will be repeated or resumed upon repair of the associated equipment of 4.3.5.1 at the discretion of the Government inspector or naval test activity.

4.5.6 Spare parts. - Spare parts furnished with an inspection lot (see 4.1.2.2) will be accepted without firing tests provided like parts assembled in the feed mechanisms of the lot are accepted after all tests. In case of failure of any part in any acceptance test, like spare parts from the same lot as the defective components shall be accepted only after screening by the contractor for the revealed defect and resubmitting for inspection by the Government.

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4.5.6.1 Replacement parts. - Each lot of replacement parts (see 4.1.2.3) will ordinarily be accepted on the basis of the inspection and test of 4.2.2; however, the Government inspector may, if he deems it necessary, submit a representative sample of replacement parts to a test activity, to be designated by the Bureau of Ordnance, for firing or other tests as required.

4.5.7 Attributes not normally considered. - Attributes not normally considered for acceptance or rejection shall be subject to corrective action. Failure on the part of the contractor to take corrective action at a time satisfactory to the Government inspector shall be sufficient cause for rejection of all lots containing such defects processed thereafter.

4.6 Rejected lots. - Rejected parts and assemblies will be returned to the contractor. Such material shall be replaced or reworked at the contractor's expense, and resubmitted for Government inspection and test. Such parts and assemblies shall be subjected to complete inspections and tests as for parts and assemblies originally submitted. A rejected lot shall not be resubmitted more than once without prior approval of the procuring activity.

4.6.1 Costs of firing tests. - In case of rejection after firing tests, the cost of such firing tests shall be borne by the contractor.

5. PREPARATION FOR DELIVERY.

5.1 General. - After all acceptance inspection and tests have been completed, the feed mechanisms shall be disassembled and prepared for delivery as hereinafter specified.

5.2 Cleaning and drying. -

5.2.1 All feed mechanisms shall be disassembled for cleaning. All parts of the feed mechanisms except the feed switch subassembly, buffer washers, "T" section seals, "O" rings, back up rings, and contact button bushing shall be cleaned by immersing the parts in a primary tank of solvent conforming to Specification P-S-661, Type 1, and draining. Immerse in a second tank of solvent as specified for the first immersion and drain. The parts are then immersed in corrosion preventive, finger print remover compound conforming to Specification MIL-C-15074 and drained. The parts are then immersed in solvent conforming to Specification P-S-661, Type 1, and thoroughly rinsed in this solvent to insure that all finger print remover compound is removed from all surfaces. However, parts, with the exceptions noted above, of mechanisms that have been subjected to firing tests, shall first be cleaned by scrubbing or wiping with a

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brush or clean cloth soaked in rifle bore cleaner conforming to Specification JAN-C-372. All parts, except the drive housing and buffer plate shall be dried using compressed air, drying ovens, infra-red lamps, or by wiping with clean cloths. The drive housing and buffer plate shall be dried with compressed air. Direct the air through the air passages in the drive housing and buffer plate to remove solvent and any foreign matter.

5.2.2 Buffer washers, etc. - Buffer washers, "T" section seals, "O"-rings, back up rings, and contact button bushing shall be cleaned in a soap solution prepared by adding one-quarter pound of soap chips conforming to Specification P-S-566 to one gallon of fresh water. After cleaning, these components shall be thoroughly rinsed in fresh water and dried by wiping with a clean cloth.

5.2.3 Feed switch subassemblies. - Feed switch subassemblies shall be cleaned by wiping with a clean dry cloth and shall not in any way be dampened with cleaning solvent.

5.3 Assembly and lubrication. - Assemble each feed mechanism and lubricate in accordance with Publication OP 1912 and 3.11. Following assembly and lubrication, test operate each mechanism by subjecting it to the functional cycling tests of 4.3.3.

5.4 Preservation and packaging. -

5.4.1 Level "A". - Following the test operation of 5.3, wipe all exposed surfaces of the feed mechanism with a clean cloth and preserve and package the feed mechanisms in accordance with the instructions set forth on drawing LD 256076.

5.4.2 Level "B". - Same as Level "A" (see 5.4.1).

5.4.3 Level "C". - Each feed mechanism shall be preserved and packaged in accordance with the manufacturer's commercial practice to assure adequate protection to prevent corrosion, deterioration, and physical damage during shipment from the manufacturer to the first receiving activity for immediate use.

5.5 Packing. -

5.5.1 Level "A". - The placing and securing of each feed mechanism in a container as specified in 5.4.1 shall constitute packing for shipment.

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5.5.2 Level "B". - Same as Level "A" (see 5.5.1).

5.5.3 Level "C". - Feed mechanisms packaged as specified in the contract or order shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.6 Spare parts. - Spare parts shall be cleaned, preserved, packaged, and packed as specified in the contract or order (see 6.2).

5.7 Marking. - In addition to any special marking required by the contract or order (see 6.2), shipping containers (see 5.4.1) shall be marked in accordance with Standard MIL-STD-129 and with the serial number of the feed mechanism.

6. NOTES

6.1 Intended use. - The 20MM Feed Mechanism Mark 7 Mods 0 and 2 are intended for use in connection with the 20MM Aircraft Guns Mark 12 Mods 0 and 2, 20MM Ammunition Links Mark 2 Mods 1 and 2, and 20MM Ammunition for Gun Mark 12 Mods 0 and 2.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, date, and number of this specification.
(b) Mark and Mods. required (see 1.2).
(c) Number of feed mechanisms assembled to each hand of feed (see 3.12).
(d) Spare parts required (see 3.7).
(e) Government property to be loaned or furnished for the conductance of tests (see 3.13, 3.14, and 4.3).
(f) Place of inspection and tests if different from 4.2.2.
(g) If preproduction samples are not required (see 4.1.1.1).
(h) Parts replacement schedule (see 4.3.5.6.1).
(i) Selection of applicable levels of preservation and packaging, and packing.
(j) Cleaning, preserving, packaging, and packing of spare parts (see 5.6).
(k) Marking requirements if different from 5.7.

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6.3 Adherence to drawings and specifications. - The contractor is required by provisions of this specification to adhere rigidly to the drawings listed in 1.2. In the event of conflict between the drawings and the performance and test requirements specified herein, the contractor shall promptly notify the procuring activity of such fact in order that appropriate changes to the applicable drawings, parts, or performance requirements may be considered by the procuring activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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